

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 6
 - More on Open Sets
 - Closed Sets

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 - More on Open Sets
 - Closed Sets

Warm-up!

Problem

Let's prove that an open ball is an open set.

Unions of open sets are open

Theorem (Open Union Theorem)

*Suppose that $\{U_i : i \in I\}$ is an arbitrary family of open sets.
Then $\bigcup_{i \in I} U_i$ is open.*

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Then there exists $j \in I$ with $\vec{x} \in U_j$.

Since U_j is open, this means $\vec{x}$ is an interior point of U_j .

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Therefore there exists $r > 0$ such that $B(\vec{x}; r) \subseteq U_j$.

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Thus $\vec{x}$ is an interior point of $\bigcup_{i \in I} U_i$.

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Thus $\vec{x}$ is an interior point of $\bigcup_{i \in I} U_i$.

Since $\vec{x}$ is arbitrary, this proves that $\bigcup_{i \in I} U_i$ is open. □

Finite intersections of open sets are open

Theorem (Open Intersection Theorem)

Suppose that $U_1, U_2, \dots, U_n$ are open sets. Then $\bigcap_{i=1}^n U_i$ is open.

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Then for all i , $\vec{x} \in U_i$.

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Take $r = \min\{r_i : 1 \leq i \leq n\}$.

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Therefore $B(\vec{x}; r) \subseteq \bigcap_{i=1}^n U_i$.

Thus $\vec{x}$ is an interior point of $\bigcap_{i=1}^n U_i$.

Since $\vec{x}$ is arbitrary, this proves that $\bigcap_{i=1}^n U_i$ is open. □

Challenge

Problem

Show that the infinite intersection

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right)$$

is not open.

Component intervals

Let $U \subseteq \mathbb{R}$ be open.

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- $(-\infty, 0)$ is also a component interval of $\mathbb{R} \setminus \{0, 1, 2, 3\}$
- we will show all open sets of $\mathbb{R}$ are made of component intervals!

Component intervals

Lemma

If I_1 and I_2 are two component intervals of an open subset $U \subseteq \mathbb{R}$, then $I_1 = I_2$ or $I_1 \cap I_2 = \emptyset$.

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Suppose that $I_1 \cap I_2 \neq \emptyset$.

Then $J := I_1 \cup I_2$ is an interval.

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Also $I_1 \subseteq U$ and $I_2 \subseteq U$, so $J \subseteq U$.

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Since I_i is a component interval and $I_i \subseteq J \subseteq U$, we have $I_i = J$.

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Since I_i is a component interval and $I_i \subseteq J \subseteq U$, we have $I_i = J$.

In particular $I_1 = J = I_2$. □

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Theorem (Apostol 3.10)

If $U \subseteq \mathbb{R}$ is open and $x \in U$, then there is a unique component interval of U containing x .

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If A is not bounded below, let $a = -\infty$. Otherwise, let $a = \inf(A)$.

If B is not bounded above, let $b = \infty$. Otherwise, let $b = \sup(B)$.

Claim: (a, b) is a component interval of U containing x . $\square$

Challenge

Problem

Prove that (a, b) is a component interval of U .

Open subsets of $\mathbb{R}$

Theorem (Representation Theorem for Open Sets in $\mathbb{R}$)

If $U \subseteq \mathbb{R}$ is open, then U is the union of a countable family of disjoint open intervals.

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Uh oh ... this isn't a **countable** union.



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Instead, we consider rational points $U \cap \mathbb{Q}$.

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Any interval contains a rational number $r \in I_x$.

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Thus $x \in \bigcup_{r \in U \cap \mathbb{Q}} I_r \subseteq U$.

Since x was arbitrary,

$$U = \bigcup_{r \in U \cap \mathbb{Q}} I_r$$

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Examples:

- singleton sets!

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Examples:

- singleton sets!
- products of closed intervals

Challenge!

Problem

Prove that a singleton set

$$A = \{\vec{a}\}$$

in $\mathbb{R}^n$ is closed.

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Solution

We must show $U = \mathbb{R}^n \setminus \{\vec{a}\}$ is open.

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Solution

We must show $U = \mathbb{R}^n \setminus \{\vec{a}\}$ is open.

Suppose $\vec{x} \in U$.

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Solution

We must show $U = \mathbb{R}^n \setminus \{\vec{a}\}$ is open.

Suppose $\vec{x} \in U$. We must show $\vec{x}$ is an interior point of U .

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Suppose $\vec{x} \in U$. We must show $\vec{x}$ is an interior point of U .

Take $r = |\vec{x} - \vec{a}|$.

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Take $r = |\vec{x} - \vec{a}|$. Then $B(\vec{x}; r)$ does not contain $\vec{a}$.

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Therefore $B(\vec{x}; r) \subseteq U$, so that $\vec{x}$ is an interior point.

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Therefore $B(\vec{x}; r) \subseteq U$, so that $\vec{x}$ is an interior point.

Since $\vec{x}$ was arbitrary, this proves U is open.

Challenge!

Problem

Prove that the **closed square**

$$[a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}$$

is a closed set.

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$$((-\infty, a) \times \mathbb{R}) \cup ((b, \infty) \times \mathbb{R}) \cup (\mathbb{R} \times (-\infty, c)) \cup (\mathbb{R} \times (d, \infty))$$

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Union of open sets is open!

Adherent and Accumulation Points

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It is called an **accumulation point** if for all $r > 0$ the ball $B(\vec{x}; r)$ contains at least one element of A **different from $\vec{x}$** .

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It is called an **accumulation point** if for all $r > 0$ the ball $B(\vec{x}; r)$ contains at least one element of A **different from $\vec{x}$** .

Examples:

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- suprema and infima are accumulation points!
- 0 is an accumulation point of $\{1/1, 1/2, 1/3, \dots\}$